

Supplementary Appendix

Causal Emergence in U.S. Equity Industry Portfolios: Dynamic Organization and Effective Dimensionality During Systemic Stress

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1 Scale-Invariance Numerical Test and Formal Derivation

A central property of the empirical estimator developed in this paper is exact invariance to common/global scalar rescaling of asset return units ($X \rightarrow cX$ for $c > 0$). This ensures that whether returns are expressed in decimals, percentages, or basis points, the information-theoretic quantities CEFI_t and q_t^* remain numerically identical.

1.1 Mathematical Derivation of Scale Invariance

Consider the transition dynamics $\mathbf{x}_{t+1} = \mathbf{A}\mathbf{x}_t + \boldsymbol{\varepsilon}_{t+1}$ with $\boldsymbol{\varepsilon}_{t+1} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_\varepsilon)$ and unconditional state covariance $\boldsymbol{\Sigma}_x = \text{Cov}(\mathbf{x}_t)$. Under a global scalar change of return units $\tilde{\mathbf{x}}_t = c\mathbf{x}_t$ ($c > 0$):

1. **State Covariance:** $\tilde{\boldsymbol{\Sigma}}_x = \text{Cov}(c\mathbf{x}_t) = c^2\boldsymbol{\Sigma}_x$.
2. **Transition Matrix Estimator:** Under the trace-scaled ridge estimator,

$$\tilde{\lambda}_t = \lambda_0 \cdot \frac{\text{Tr}(\tilde{\mathbf{X}}_{\text{lag}}^\top \tilde{\mathbf{X}}_{\text{lag}})}{p} = c^2\lambda_t \quad (1)$$

Consequently,

$$\hat{\tilde{\mathbf{A}}} = \tilde{\mathbf{X}}_{\text{lead}}^\top \tilde{\mathbf{X}}_{\text{lag}} \left(\tilde{\mathbf{X}}_{\text{lag}}^\top \tilde{\mathbf{X}}_{\text{lag}} + \tilde{\lambda}_t \mathbf{I}_p \right)^{-1} = c^2 \mathbf{X}_{\text{lead}}^\top \mathbf{X}_{\text{lag}} \left(c^2 \mathbf{X}_{\text{lag}}^\top \mathbf{X}_{\text{lag}} + c^2 \lambda_t \mathbf{I}_p \right)^{-1} = \hat{\mathbf{A}} \quad (2)$$

Thus, $\hat{\mathbf{A}}$ is strictly scale-invariant.

3. **Innovation Covariance:** Residuals transform as $\tilde{\boldsymbol{\varepsilon}}_t = c\boldsymbol{\varepsilon}_t$, so under Ledoit-Wolf analytical shrinkage, $\tilde{\boldsymbol{\Sigma}}_\varepsilon = c^2\boldsymbol{\Sigma}_\varepsilon$.
4. **Intervention Scale:** The energy-scaled variance is:

$$\tilde{\sigma}_{do,t}^2 = \kappa^2 \cdot \frac{\text{Tr}(\tilde{\boldsymbol{\Sigma}}_{x,t})}{p} = c^2 \sigma_{do,t}^2 \quad (3)$$

5. **Effective Information and Whitened SPD Representation:** In whitened coordinates, the micro-transition covariance is represented by the symmetric positive definite (SPD) matrix:

$$\mathbf{M}_{\text{micro}} = \mathbf{I}_p + \sigma_{do,t}^2 \boldsymbol{\Sigma}_\varepsilon^{-1/2} \hat{\mathbf{A}} \hat{\mathbf{A}}^\top \boldsymbol{\Sigma}_\varepsilon^{-1/2} \quad (4)$$

By Sylvester's determinant theorem ($\det(\mathbf{I} + \mathbf{AB}) = \det(\mathbf{I} + \mathbf{BA})$), $\det(\mathbf{M}_{\text{micro}}) = \det(\mathbf{I}_p + \sigma_{do,t}^2 \hat{\mathbf{A}} \hat{\mathbf{A}}^\top \Sigma_\varepsilon^{-1})$. Under global rescaling $\tilde{\mathbf{x}}_t = c\mathbf{x}_t$:

$$\begin{aligned} \widetilde{EI}(\mathbf{x}) &= \frac{1}{2} \ln \det \left(\mathbf{I}_p + \tilde{\sigma}_{do,t}^2 \Sigma_\varepsilon^{-1/2} \hat{\mathbf{A}} \hat{\mathbf{A}}^\top \tilde{\Sigma}_\varepsilon^{-1/2} \right) \\ &= \frac{1}{2} \ln \det \left(\mathbf{I}_p + (c^2 \sigma_{do,t}^2) (c^2 \Sigma_\varepsilon)^{-1/2} \hat{\mathbf{A}} \hat{\mathbf{A}}^\top (c^2 \Sigma_\varepsilon)^{-1/2} \right) \\ &= \frac{1}{2} \ln \det (\mathbf{M}_{\text{micro}}) = EI(\mathbf{x}) \end{aligned} \quad (5)$$

Numerical computation is executed via the Cholesky factor of the shrunk innovation covariance $\Sigma_\varepsilon = \mathbf{L}\mathbf{L}^\top$, setting $\mathbf{Y} = \mathbf{L}^{-1} \hat{\mathbf{A}}$ and $\mathbf{M}_{\text{micro}} = \mathbf{I}_p + \sigma_{do,t}^2 \mathbf{Y}\mathbf{Y}^\top$, guaranteeing strict positive definiteness and numerical stability.

Identical algebraic cancellations hold for any projected macro-system $EI_q(\mathbf{W})$, guaranteeing that CEFI_t and q_t^* (evaluated in nats/DOF) are strictly scale-invariant.

Scope Delimitation. Invariance is claimed exclusively for common scalar multipliers ($X \rightarrow cX$). Invariance is not claimed for arbitrary asset-specific diagonal scalings ($X \rightarrow XD$).

Table 1: Numerical Verification of Global Return Scale Invariance Across Four Orders of Magnitude

Scale Multiplier (c)	EI_{micro}	$\text{EI}_{\text{macro}}^*$	CEFI	q^*	$ \Delta \text{CEFI} $
$c = 0.01$ (Basis points / 100)	0.11381190	0.02644265	-0.00022650	7	0.00
$c = 1.00$ (Standard return units)	0.11381190	0.02644265	-0.00022650	7	$< 10^{-14}$
$c = 100.0$ (Percentage points)	0.11381190	0.02644265	-0.00022650	7	$< 10^{-14}$
$c = 10000.0$ (Basis points)	0.11381190	0.02644265	-0.00022650	7	$< 10^{-14}$

2 VAR Transition Matrix Orientation: Mathematical Derivation and Synthetic Test

Let $\mathbf{x}_t \in \mathbb{R}^p$ denote a column vector of asset returns at date t . The theoretical first-order Vector Autoregression is defined as:

$$\mathbf{x}_{t+1} = \mathbf{A}\mathbf{x}_t + \boldsymbol{\varepsilon}_{t+1}, \quad \boldsymbol{\varepsilon}_{t+1} \sim \mathcal{N}(\mathbf{0}, \Sigma_\varepsilon) \quad (6)$$

where $\mathbf{A} \in \mathbb{R}^{p \times p}$ maps state $\mathbf{x}_t$ to the conditional expectation $\mathbb{E}[\mathbf{x}_{t+1} \mid \mathbf{x}_t] = \mathbf{A}\mathbf{x}_t$.

In matrix data arrangements, observations are organized into matrices $\mathbf{X}_{\text{lag}} \in \mathbb{R}^{(T-1) \times p}$ and $\mathbf{X}_{\text{lead}} \in \mathbb{R}^{(T-1) \times p}$, where row t contains $\mathbf{x}_t^\top$ and $\mathbf{x}_{t+1}^\top$, respectively. Transposing the model equation yields the row-regression representation:

$$\mathbf{x}_{t+1}^\top = \mathbf{x}_t^\top \mathbf{A}^\top + \boldsymbol{\varepsilon}_{t+1}^\top \implies \mathbf{X}_{\text{lead}} = \mathbf{X}_{\text{lag}} \mathbf{B} + \mathbf{E}, \quad \mathbf{B} = \mathbf{A}^\top \quad (7)$$

The least-squares / ridge solution for $\mathbf{B}$ is:

$$\hat{\mathbf{B}} = (\mathbf{X}_{\text{lag}}^\top \mathbf{X}_{\text{lag}} + \lambda_t \mathbf{I}_p)^{-1} \mathbf{X}_{\text{lag}}^\top \mathbf{X}_{\text{lead}} = \hat{\mathbf{A}}^\top \quad (8)$$

Taking the transpose recovers the column transition matrix:

$$\hat{\mathbf{A}} = \hat{\mathbf{B}}^\top = \mathbf{X}_{\text{lead}}^\top \mathbf{X}_{\text{lag}} (\mathbf{X}_{\text{lag}}^\top \mathbf{X}_{\text{lag}} + \lambda_t \mathbf{I}_p)^{-1} \quad (9)$$

2.1 Synthetic Non-Symmetric Recovery Audit

To verify that the implementation receives $\mathbf{A}$ and not $\mathbf{A}^\top$, we generated $T = 10,000$ synthetic observations from a strongly non-symmetric transition matrix $\mathbf{A}_{\text{true}}$ ($\|\mathbf{A}_{\text{true}} - \mathbf{A}_{\text{true}}^\top\|_F = 0.738$, spectral radius $\rho = 0.528$). Fitting via our estimator yields:

$$\|\hat{\mathbf{A}} - \mathbf{A}_{\text{true}}\|_F = 0.061089, \quad \|\hat{\mathbf{A}}^\top - \mathbf{A}_{\text{true}}\|_F = 1.589664 \quad (10)$$

The direct estimation error is over 26 times smaller than the transposed error, confirming that the estimator recovers the correctly oriented column-model transition matrix $\mathbf{A}$, rather than $\mathbf{A}^\top$.

3 Macro-Dynamics: Interventional Lifting and Observational Closure Diagnostic

Let $\mathbf{W} \in \mathbb{R}^{q \times p}$ denote an orthogonal coarse-graining matrix residing on the row-Stiefel manifold $\mathcal{V}_q(\mathbb{R}^p) = \{\mathbf{W} \in \mathbb{R}^{q \times p} : \mathbf{W}\mathbf{W}^\top = \mathbf{I}_q\}$. The macroscopic state is defined as $\mathbf{y}_t = \mathbf{W}\mathbf{x}_t$.

3.1 Interventional Channel Construction via Canonical Lifting

Following the causal emergence framework of [Liu et al. \(2024\)](#), an intervention on the macro-state $do(\mathbf{y}_t)$ is lifted to the micro-state via the right pseudo-inverse $do(\mathbf{x}_t) = \mathbf{W}^\dagger do(\mathbf{y}_t)$.

Because $\mathbf{W}\mathbf{W}^\top = \mathbf{I}_q$, the right inverse is $\mathbf{W}^\dagger = \mathbf{W}^\top(\mathbf{W}\mathbf{W}^\top)^{-1} = \mathbf{W}^\top$.

Under the interventional lifting $do(\mathbf{x}_t) = \mathbf{W}^\top \mathbf{y}_t$, the micro-state evolves as:

$$\mathbf{x}_{t+1} \mid do(\mathbf{y}_t) = \mathbf{A}\mathbf{W}^\top \mathbf{y}_t + \boldsymbol{\varepsilon}_{t+1} \quad (11)$$

Projecting the resulting state back into the macro-subspace via $\mathbf{W}$ defines the *constructed macro interventional channel*:

$$\mathbf{y}_{t+1} = \mathbf{W}\mathbf{x}_{t+1} = (\mathbf{W}\mathbf{A}\mathbf{W}^\top)\mathbf{y}_t + \mathbf{W}\boldsymbol{\varepsilon}_{t+1} = \mathbf{A}_M\mathbf{y}_t + \boldsymbol{\varepsilon}_{M,t+1} \quad (12)$$

where $\mathbf{A}_M = \mathbf{W}\mathbf{A}\mathbf{W}^\top$ and $\boldsymbol{\Sigma}_M = \mathbf{W}\boldsymbol{\Sigma}_\varepsilon\mathbf{W}^\top$. Defining the whitened macro transition matrix:

$$\mathbf{M}_M(\mathbf{W}) = \mathbf{I}_q + \sigma_{do,t}^2 \boldsymbol{\Sigma}_M^{-1/2} \mathbf{A}_M \mathbf{A}_M^\top \boldsymbol{\Sigma}_M^{-1/2} \quad (13)$$

By Sylvester's determinant identity, $\det(\mathbf{M}_M(\mathbf{W})) = \det(\mathbf{I}_q + \sigma_{do,t}^2 \mathbf{A}_M \mathbf{A}_M^\top \boldsymbol{\Sigma}_M^{-1})$. The continuous macro Effective Information is evaluated as:

$$EI_q(\mathbf{W}) = \frac{1}{2} \ln \det(\mathbf{M}_M(\mathbf{W})) = \frac{1}{2} \ln \det(\mathbf{I}_q + \sigma_{do,t}^2 \mathbf{A}_M \mathbf{A}_M^\top \boldsymbol{\Sigma}_M^{-1}) \quad (14)$$

where numerical stability is preserved via the Cholesky factor $\boldsymbol{\Sigma}_M = \mathbf{L}_M \mathbf{L}_M^\top$, setting $\mathbf{Y}_M = \mathbf{L}_M^{-1} \mathbf{A}_M$ and $\mathbf{M}_M = \mathbf{I}_q + \sigma_{do,t}^2 \mathbf{Y}_M \mathbf{Y}_M^\top$.

3.2 Observational Projection and Closure Error Diagnostic

Under passive observational dynamics, the true projected process is:

$$\mathbf{y}_{t+1} = \mathbf{W}\mathbf{x}_{t+1} = \mathbf{W}\mathbf{A}\mathbf{x}_t + \mathbf{W}\boldsymbol{\varepsilon}_{t+1} = \mathbf{W}\mathbf{A}\mathbf{W}^\top \mathbf{y}_t + \mathbf{W}\mathbf{A}(\mathbf{I}_p - \mathbf{W}^\top \mathbf{W})\mathbf{x}_t + \mathbf{W}\boldsymbol{\varepsilon}_{t+1} \quad (15)$$

The middle term represents omitted micro-state dynamics orthogonal to the macro-subspace. We define the relative observational closure error diagnostic as:

$$r_{\text{closure},t} = \frac{\|\mathbf{W}\mathbf{A} - \mathbf{W}\mathbf{A}\mathbf{W}^\top \mathbf{W}\|_F}{\|\mathbf{W}\mathbf{A}\|_F} \quad (16)$$

Because r_{closure} is strictly positive ($0.39 \leq r_{\text{closure}} \leq 0.48$), the macro model $\mathbf{A}_M = \mathbf{W}\mathbf{A}\mathbf{W}^\top$ is formally interpreted as a *constructed macro interventional transition operator* under the canonical lifting $\mathbf{W}^\top$, rather than an autonomous closed observational projection.

Table 2: Observational Closure Diagnostics at Optimal Macro-Dimension q^*

Market Episode	Benchmark Date	Optimal q^*	Closure Ratio r_{closure}	Observational
2005 Calm Benchmark	2005-12-30	1	0.3915	Structurally
2008 GFC Peak	2008-11-20	1	0.4350	Structurally
2020 COVID Crash	2020-03-23	2	0.4826	Structurally
2000 Dot-Com Crash	2001-03-20	6	0.4560	Structurally
2022 Rate Tightening	2022-06-15	1	0.4637	Structurally

Notes: The non-zero closure ratios ($r_{\text{closure}} \in [0.39, 0.48] > 0$) indicate that the optimized macro representations are not observationally closed; $\mathbf{WAW}^\top$ is therefore interpreted throughout as the transition operator of the constructed interventional macro channel rather than an autonomous observational system.

4 Optimizer Budget Sensitivity Diagnostic

4.1 Riemannian Gradient and Canonical Metric Duality on Row-Stiefel

The row-Stiefel manifold is defined as $\mathcal{V}_q(\mathbb{R}^p) = \{\mathbf{W} \in \mathbb{R}^{q \times p} : \mathbf{W}\mathbf{W}^\top = \mathbf{I}_q\}$. The canonical metric on $T_{\mathbf{W}}\mathcal{V}_q(\mathbb{R}^p)$ is defined by [Edelman et al. \(1998\)](#) and [Absil et al. \(2008\)](#):

$$g_{\mathbf{W}}^{\text{canonical}}(\Delta_1, \Delta_2) = \text{Tr} \left(\Delta_1 \left(\mathbf{I}_p - \frac{1}{2} \mathbf{W}^\top \mathbf{W} \right) \Delta_2^\top \right) \quad (17)$$

Under this metric, the Riemannian gradient of a differentiable scalar function $f(\mathbf{W})$ with Euclidean gradient $\mathbf{G} = \nabla_{\mathbf{W}} f(\mathbf{W})$ is:

$$\text{grad}_{\mathcal{R}} f(\mathbf{W}) = \mathbf{G} - \mathbf{W}\mathbf{G}^\top \mathbf{W} \quad (18)$$

Theorem (Canonical Metric Inner Product Duality). For any tangent vector $\Delta \in T_{\mathbf{W}}\mathcal{V}_q(\mathbb{R}^p)$ (which satisfies $\mathbf{W}\Delta^\top + \Delta\mathbf{W}^\top = \mathbf{0}$):

$$g_{\mathbf{W}}^{\text{canonical}}(\text{grad}_{\mathcal{R}} f(\mathbf{W}), \Delta) = \text{Tr}(\mathbf{G}\Delta^\top) = \langle \mathbf{G}, \Delta \rangle_{\text{Euclidean}} = Df(\mathbf{W})[\Delta] \quad (19)$$

4.2 Optimizer Budget Sensitivity and Calibration Diagnostic

We evaluated 25 evenly spaced historical estimation windows comparing alternative computational budgets against a high-budget reference configuration (150 iterations, 25 multistarts). The objective gap is evaluated on the true dimension-selection criterion $J(q^*) = \frac{EI_{q^*}}{q^*} - \frac{EI_p}{p}$.

Increasing the search budget generally reduces the objective-value gap relative to the high-budget reference. At 12 multistarts and 100 iterations, the median objective gap drops

Table 3: Optimizer Budget Calibration Diagnostic on Selection Objective $J(q^*)$: Candidate Configurations vs. High-Budget Reference (25/150)

Configuration	Restarts	Iter	Cost Mult.	Median Gap (%)	Q95 Gap (%)	Pearson ρ	Exact q^*	$q^* \pm 1$
4/35 (Calibration baseline)	4	35	1.00×	25.08%	40.74%	0.8452	52.0%	88.0%
8/75	8	75	2.90×	5.31%	16.84%	0.9533	72.0%	84.0%
12/100 [Production]	12	100	5.90×	1.82%	11.70%	0.9732	92.0%	100.0%
16/100	16	100	7.84×	1.20%	8.39%	0.9756	88.0%	92.0%
25/150 (Reference)	25	150	18.02×	0.00%	0.00%	1.0000	100.0%	100.0%

to 1.82%, with 92.0% exact dimensional agreement with the high-budget reference and 100.0% agreement within ± 1 dimension (Pearson $\rho = 0.9732$, Spearman $\rho_S = 0.9377$). Because observed and surrogate null series are evaluated under identical optimizer budgets, Monte Carlo hypothesis tests avoid differential optimization-budget bias.

5 VAR Stability and Spectral Radius Diagnostics

We evaluate the spectral radius $\rho_t = \max_j |\lambda_j(\mathbf{A}_t)|$ across all 4,346 rolling windows (1992–2026). The empirical distribution yields:

- Mean $\rho_t = 0.3207$, Median $\rho_t = 0.3078$, $Q_{95} = 0.4227$, Maximum $\rho_t = 0.6777$.
- Zero rolling windows exhibit $\rho_t \geq 1.0$, confirming that all estimated VAR(1) transition operators are strictly stationary.

Table 4: Spectral Radius of VAR(1) Transition Matrix Across Historical Stress Episodes

Historical Episode	Mean ρ_t	Median ρ_t	Max ρ_t	Stationary (%)
2005 Calm Market Benchmark	0.2948	0.2910	0.3340	100%
2008 GFC Peak	0.3159	0.3120	0.4120	100%
2020 COVID Shock	0.4222	0.4180	0.5890	100%
2000 Dot-Com Crash	0.3024	0.2990	0.3610	100%
2022 Rate Tightening	0.3015	0.2980	0.3580	100%

6 VAR Dynamic Specification and Lag Adequacy Diagnostics

The empirical design adopts a first-order Vector Autoregression as the deliberate specification for capturing one-step intertemporal transfer capacity between consecutive trading days. Residual diagnostic tests on the estimated innovation vector $\boldsymbol{\varepsilon}_t = \mathbf{x}_t - \hat{\mathbf{A}}\mathbf{x}_{t-1}$ show:

- **Autocorrelation:** Mean lag-1 residual autocorrelation across assets is near zero in calm markets ($\bar{r}_1 = +0.0064$), GFC ($\bar{r}_1 = -0.0454$), and COVID ($\bar{r}_1 = +0.0097$).
- **Volatility Clustering:** As expected in daily financial returns, innovations exhibit ARCH effects and fat tails (mean excess kurtosis ranging from +0.74 in 2005 to +6.64 in March 2020).

7 Full Matched Null Hierarchy Simulation Protocol and Results

7.1 Surrogate Generation and Estimation Protocol

For each historical benchmark estimation window $\mathbf{X} \in \mathbb{R}^{W \times p}$ ($W = 500$ trading days, $p = 30$ industry portfolios), surrogate null time series ensembles are generated and evaluated under the following unified protocol:

1. H_0^{circ} (**Circular Shift Null**): Each portfolio return column is independently rolled by a random integer lag $\tau_j \in [10, W - 10]$ with periodic boundary wrapping, destroying contemporaneous and cross-lag correlations while preserving marginal autocorrelation.
2. H_0^{diag} (**Decoupled Diagonal VAR Null**): Simulates uncoupled AR(1) processes $\mathbf{x}_{t+1} = \text{diag}(\hat{\mathbf{A}})\mathbf{x}_t + \boldsymbol{\varepsilon}_{t+1}$ with independent Gaussian innovations $\boldsymbol{\varepsilon}_{t+1} \sim \mathcal{N}(\mathbf{0}, \text{diag}(\boldsymbol{\Sigma}_\varepsilon))$. Initial state is drawn from the innovation distribution, initialized with a 50-step burn-in to reach empirical stationarity, followed by $W = 500$ simulated observations.
3. H_0^{static} (**Static Correlation Mode Null**): Synchronously permutes time steps across all assets via a uniform random permutation $\pi \in \mathcal{S}_W$, preserving the exact instantaneous covariance $\boldsymbol{\Sigma}_x$ while destroying all intertemporal causal transitions $\mathbf{x}_t \rightarrow \mathbf{x}_{t+1}$.
4. $H_0^{\text{diag+contemp}}$ (**Cross-Lag Network Isolation Null**): Simulates $\mathbf{x}_{t+1} = \text{diag}(\hat{\mathbf{A}})\mathbf{x}_t + \boldsymbol{\varepsilon}_{t+1}$ driven by the full empirical contemporaneous innovation covariance $\boldsymbol{\Sigma}_\varepsilon$ via Cholesky factor $\mathbf{L}$ (with regularizing trace jitter $10^{-10} \text{Tr}(\boldsymbol{\Sigma}_\varepsilon)/p$). Follows a 50-step burn-in followed by $W = 500$ simulated steps, isolating own-lag persistence and instantaneous shock co-movement while setting cross-lag coupling to zero ($A_{ij} = 0, \forall i \neq j$).

Every surrogate realization $\mathbf{X}_{\text{null}} \in \mathbb{R}^{W \times p}$ is passed through the complete estimation pipeline: $\hat{\mathbf{A}}_{\text{null}}$ is fitted via trace-scaled ridge regression ($\lambda_0 = 10^{-4}$), $\boldsymbol{\Sigma}_{\varepsilon, \text{null}}$ via Ledoit-Wolf shrinkage, and $\sigma_{do, \text{null}}^2 = \kappa^2 \text{Tr}(\boldsymbol{\Sigma}_{x, \text{null}})/p$. The Riemannian Stiefel optimization is executed for all candidate dimensions $q \in \{1, \dots, p - 1\}$ under the identical optimizer

budget as observed data (12 restarts / 100 iterations), ensuring strictly symmetric Monte Carlo hypothesis testing. Reproducibility is governed by a deterministic seed schedule (seed = 42 + regime.id $\times$ 10⁵ + null.id $\times$ 10⁴ + b).

7.2 Matched Null Inference Results

Table 5: Complete Matched Null Model Inference Table ($B = 9,999$ for Primary Nulls; $B = 999$ for Auxiliary Nulls)

Regime	Null Model	B	CEFI _{obs}	$\mathbb{E}[\text{CEFI}_0]$	Q ₉₅	z _{dev}	P _{emp}	P _{Holm}
Calm Period (2005)	H_0^{circ} (Circular)	999	0.2023	+0.0708	+0.2094	+1.94	0.0630	–
	H_0^{diag} (VAR Diag)	999	0.2023	+0.0128	+0.0390	+15.05	0.0010	–
	H_0^{static} (Static Mode)	9,999	0.2023	+0.1500	+0.2253	+1.26	0.1160	0.3480
	$H_0^{\text{diag+contemp}}$ (Cross-lag Isol.)	9,999	0.2023	+0.1654	+0.2504	+0.80	0.2059	0.4118
2008 GFC Peak	H_0^{circ} (Circular)	999	0.3656	+0.1808	+0.3642	+1.56	0.0490	–
	H_0^{diag} (VAR Diag)	999	0.3656	+0.0356	+0.0691	+18.08	0.0010	–
	H_0^{static} (Static Mode)	9,999	0.3656	+0.2774	+0.3714	+1.70	0.0587	0.2860
	$H_0^{\text{diag+contemp}}$ (Cross-lag Isol.)	9,999	0.3656	+0.2878	+0.3695	+1.70	0.0572	0.2860
2020 COVID Shock	H_0^{circ} (Circular)	999	0.3342	+0.1466	+0.2640	+2.90	0.0040	–
	H_0^{diag} (VAR Diag)	999	0.3342	+0.0488	+0.0927	+11.54	0.0010	–
	H_0^{static} (Static Mode)	9,999	0.3342	+0.2213	+0.3070	+2.40	0.0182	0.1092
	$H_0^{\text{diag+contemp}}$ (Cross-lag Isol.)	9,999	0.3342	+0.3008	+0.4139	+0.53	0.2893	0.4118

8 Multiple-Testing Adjustments and Sensitivity Analysis

We define the **Primary Hypothesis Family** as the six central regime-null combinations:

$$\mathcal{F}_{\text{primary}} = \{(\text{Calm}, H_0^{\text{static}}), (\text{Calm}, H_0^{\text{diag+contemp}}), (\text{GFC}, H_0^{\text{static}}), (\text{GFC}, H_0^{\text{diag+contemp}}), (\text{COVID}, H_0^{\text{static}}), (\text{COVID}, H_0^{\text{diag+contemp}})\} \quad (20)$$

Under the Holm-Bonferroni step-down procedure ($m = 6$), individual nominal test rejections are evaluated alongside family-wise multiplicity-adjusted p -values.

9 Causal Effective Dimension (q^*) vs. Static Covariance Dimensionality

To evaluate whether q^* merely mirrors static covariance concentration (Effective Rank or PCA dimension), we evaluated their relationship across all 4,346 rolling windows:

- Spearman correlation between q^* and Effective Rank: $\rho_S = -0.0634$.

- Pearson correlation between q^* and Effective Rank: $\rho = -0.0781$.
- Spearman correlation between q^* and 80% PCA variance dimension: $\rho_S = -0.0312$.
- Spearman correlation between q^* and 90% PCA variance dimension: $\rho_S = -0.0450$.
- Overall historical concentration: 98.94% of observations satisfy $q^* \leq 4$ with modal $q^* = 1$ (accounting for 63.8% of all windows, and 87.7% satisfying $q^* \leq 2$).
- Cross-method dimensional concordance: analytical SVD emergence achieves 87.7% agreement within ± 1 dimension (63.2% exact dimensional match).

These low correlations indicate that q^* is only weakly monotonically associated with static covariance dimensionality metrics.

10 Conventional Systemic Risk Benchmarks: Collinearity and Residualized CEFI

10.1 Multicollinearity Diagnostics

Variance Inflation Factors (VIF) and condition number for the multivariate regression of CEFI_t on conventional proxies ($RV_t, \bar{\rho}_t, ER_t, DY_t$):

- $\text{VIF}(RV) = 3.20$, $\text{VIF}(\bar{\rho}) = 27.30$, $\text{VIF}(ER) = 13.77$, $\text{VIF}(DY) = 15.11$.
- Condition number of the normalized design matrix: $\kappa(\mathbf{X}) = 11.80$.

Because average correlation, effective rank, and connectedness exhibit substantial multicollinearity ($\text{VIF} > 13$), individual partial regression coefficients should not be interpreted as isolated economic channels. The complete linear specification accounts for 29.31% of linear variation ($R^2 = 29.31\%$, leaving 70.69% unexplained by this linear combination).

10.2 Residualized CEFI Analysis

We construct residualized CEFI:

$$\text{CEFI}_{\text{res},t} = \text{CEFI}_t - \hat{\mathbb{E}}[\text{CEFI}_t \mid RV_t, \bar{\rho}_t, ER_t, DY_t] \quad (21)$$

Evaluating $\text{CEFI}_{\text{res},t}$ across episodes:

- **2008 GFC Peak:** Mean $\text{CEFI}_{\text{res}} = -0.0025$ (Median = -0.0274).

- **2020 COVID Shock:** Mean $\text{CEFI}_{\text{res}} = +0.0017$ (Median = +0.0115).
- **2000 Dot-Com Crash:** Mean $\text{CEFI}_{\text{res}} = -0.0496$ (Median = -0.0515).
- **2022 Rate Tightening:** Mean $\text{CEFI}_{\text{res}} = +0.0941$ (Median = +0.0823).

Residualized CEFI_t retains episode-level variation after linear adjustment for conventional proxies.

11 Event Study Regressions: Full HAC Lag Bandwidth Sensitivity

Table 6: Sensitivity of Event Study Estimates Across Extended Newey-West Lag Bandwidths ($L = 20$ to $L = 250$)

HAC Lag (L)	β_{Liq}	t-stat	β_{Val}	t-stat	$\Delta\beta$	Wald t-stat	Wald p-val
$L = 20$	+0.034	+1.36	-0.071	-2.90	+0.105	+3.11	0.0019
$L = 40$	+0.034	+1.08	-0.071	-2.20	+0.105	+2.40	0.0165
$L = 60$	+0.034	+0.99	-0.071	-1.90	+0.105	+2.12	0.0336
$L = 120$	+0.034	+0.99	-0.071	-1.50	+0.105	+1.83	0.0678
$L = 250$	+0.034	+1.26	-0.071	-1.33	+0.105	+1.71	0.0878

12 Leave-One-Episode-Out Crisis Sensitivity Analysis

To verify whether the historical contrast between liquidity crises and valuation repricing ($\Delta\beta = \beta_{\text{Liq}} - \beta_{\text{Val}} > 0$) is driven by a single outlier episode, we re-estimated the specification excluding each historical episode sequentially, computing the exact contrast Wald test using the full HAC covariance matrix.

Table 7: Leave-One-Episode-Out Event Study Robustness with Full HAC Contrast Covariance

Excluded Episode	β_{Liq}	β_{Val}	$\Delta\beta$	Exact Wald t-stat	Exact Wald p-val
None (Full Sample)	+0.034	-0.071	+0.105	+2.40	0.0165
Exclude Dot-Com Crash	+0.034	+0.096	-0.063	-1.36	0.1725
Exclude 2008 GFC Peak	+0.036	-0.071	+0.107	+2.60	0.0092
Exclude 2020 COVID Shock	+0.033	-0.071	+0.105	+2.22	0.0264
Exclude 2022 Rate Tightening	+0.034	-0.136	+0.169	+5.24	1.58×10^{-7}

In single-episode exclusions omitting GFC ($\Delta\beta = +0.107, t = 2.60$), COVID ($\Delta\beta = +0.105, t = 2.22$), or 2022 tightening ($\Delta\beta = +0.169, t = 5.24$), the contrast remains strictly positive and statistically significant. However, omitting the Dot-Com crash reduces the contrast to $\Delta\beta = -0.063$ ($t = -1.36, p = 0.1725$), demonstrating that the statistical precision and magnitude of the historical regime contrast are materially influenced by the Dot-Com episode.

13 Sensitivity to Intervention Scale Parameter κ

Table 8: Sensitivity of Causal Emergence to Dimensionless Intervention Scale $\kappa \in [0.25, 4.0]$

κ	Mean CEFI	Spearman ρ_S vs Baseline	Modal q^*	β_{LiQ}	t-stat
0.25	0.0005	0.688	28	+0.0015	+1.27
0.50	0.0316	0.839	3	+0.0205	+1.04
1.00	0.2375	1.000	1	+0.0336	+1.08
2.00	0.5146	0.894	1	+0.0231	+0.68
4.00	0.7729	0.789	1	-0.0038	-0.12

14 Sensitivity to Rolling Window Length W

Table 9: Robustness Across Rolling Estimation Window Lengths ($W \in \{500, 750, 1000\}$ Trading Days)

Window Length (W)	Mean CEFI	Spearman ρ_S vs Baseline	Optimal Phase Lag (ℓ)	La
$W = 500$ days (Baseline)	0.2375	1.000	0	
$W = 750$ days	0.1676	0.819	-10 days	
$W = 1000$ days	0.1250	0.690	-140 days	

15 Cross-Partition and Industry-Granularity Sensitivity: Fama-French 49 Industry Portfolios

To evaluate whether our empirical findings are artifacts of the 30-industry partition or persist at finer sectoral granularity, we replicate the estimation pipeline on the 49 Fama-French Industry Portfolios ($p = 49$) across all $q \in \{1, \dots, 48\}$. Because this test evaluates an alternative

classification of the same underlying U.S. equity universe, it represents a sensitivity check on industry granularity and cross-partition stability rather than cross-universe or out-of-sample robustness:

- Sample period: 1990–2026 ($T = 9,190$ trading days, 4,346 rolling windows).
- Mean $\text{CEFI}_{FF49} = 0.3592$, Median $\text{CEFI}_{FF49} = 0.3483$, Modal $q^* = 1$.
- 97.13% of historical rolling windows exhibit $q^* \leq 4$.
- At the March 2020 benchmark, $\text{CEFI}_{\text{obs}} = 0.4578$; this provides an exploratory sensitivity check showing that low causal effective dimensionality and positive emergence density persist when the microscopic system is observed at finer industrial granularity.

16 Theoretical Cross-Method Benchmarking Formulations: Liu et al. (2024) and Liu et al. (2025)

16.1 Liu et al. (2024) Exact Bounded-Uniform Theory

In the exact formulation of Liu et al. (2024), causal emergence is developed for linear stochastic systems under uniform interventions over a bounded compact domain $\mathcal{D} = [-\delta, \delta]^p$:

$$\mathbf{x}_{t+1} = \mathbf{A}\mathbf{x}_t + \boldsymbol{\varepsilon}_{t+1}, \quad \boldsymbol{\varepsilon}_{t+1} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_\varepsilon), \quad do(\mathbf{x}_t) \sim \mathcal{U}([-\delta, \delta]^p) \quad (22)$$

The continuous effective information of the microscopic system is evaluated as:

$$\mathcal{J}_{\text{micro}}(\mathbf{x}) = \frac{1}{2} \ln \det(\mathbf{A}\boldsymbol{\Sigma}_x\mathbf{A}^\top + \boldsymbol{\Sigma}_\varepsilon) - \frac{1}{2} \ln \det(\boldsymbol{\Sigma}_\varepsilon) \quad (23)$$

where $\boldsymbol{\Sigma}_x$ is the state covariance. For an orthogonal projection $\mathbf{W} \in \mathcal{V}_q(\mathbb{R}^p)$ onto a q -dimensional macro-state $\mathbf{y}_t = \mathbf{W}\mathbf{x}_t$, the macro parameters are $\mathbf{A}_M = \mathbf{W}\mathbf{A}\mathbf{W}^\top$, $\boldsymbol{\Sigma}_M = \mathbf{W}\boldsymbol{\Sigma}_\varepsilon\mathbf{W}^\top$, and $\boldsymbol{\Sigma}_Y = \mathbf{W}\boldsymbol{\Sigma}_x\mathbf{W}^\top$. The macro effective information is:

$$\mathcal{J}_{\text{macro}}(\mathbf{y}) = \frac{1}{2} \ln \det(\mathbf{A}_M\boldsymbol{\Sigma}_Y\mathbf{A}_M^\top + \boldsymbol{\Sigma}_M) - \frac{1}{2} \ln \det(\boldsymbol{\Sigma}_M) \quad (24)$$

The normalized causal emergence metric per degree of freedom is defined as:

$$\Delta\mathcal{J}(\mathbf{W}) = \frac{\mathcal{J}_{\text{macro}}(\mathbf{y})}{q} - \frac{\mathcal{J}_{\text{micro}}(\mathbf{x})}{p} \quad (25)$$

and the optimal macro representation is obtained by maximizing $\Delta\mathcal{J}(\mathbf{W})$ over $\mathbf{W} \in \mathcal{V}_q(\mathbb{R}^p)$ and $q \in \{1, \dots, p-1\}$.

16.2 Liu et al. (2025) Singular Value Decomposition (SVD) Formulation

Liu et al. (2025) propose an analytical approach to causal emergence in Gaussian iterative dynamics that circumvents iterative manifold optimization. Let $\Sigma_\varepsilon = \mathbf{L}\mathbf{L}^\top$ denote the Cholesky factorization of the innovation covariance. The noise-whitened transition operator is defined as:

$$\mathbf{K} = \mathbf{L}^{-1}\mathbf{A} \in \mathbb{R}^{p \times p} \quad (26)$$

Taking the Singular Value Decomposition (SVD) of $\mathbf{K}$:

$$\mathbf{K} = \mathbf{U}\mathbf{S}\mathbf{V}^\top, \quad \mathbf{S} = \text{diag}(s_1, s_2, \dots, s_p), \quad s_1 \geq s_2 \geq \dots \geq s_p \geq 0 \quad (27)$$

The analytical coarse-graining projection for dimension q is given by the leading q left singular vectors: $\mathbf{W}_{\text{SVD}} = \mathbf{U}_{:,1:q}^\top \in \mathbb{R}^{q \times p}$. Under an isotropic Gaussian intervention with variance σ_{do}^2 , the micro and macro Effective Information evaluate in closed form directly from the singular spectrum:

$$EI_{\text{micro}} = \frac{1}{2} \sum_{i=1}^p \ln(1 + \sigma_{do}^2 s_i^2), \quad EI_{\text{macro}}(q) = \frac{1}{2} \sum_{i=1}^q \ln(1 + \sigma_{do}^2 s_i^2) \quad (28)$$

The SVD causal emergence criterion per degree of freedom is:

$$\Delta EI_{\text{SVD}}(q) = \frac{1}{2q} \sum_{i=1}^q \ln(1 + \sigma_{do}^2 s_i^2) - \frac{1}{2p} \sum_{i=1}^p \ln(1 + \sigma_{do}^2 s_i^2) \quad (29)$$

with optimal dimension $q_{\text{SVD}}^* = \text{argmax}_q \Delta EI_{\text{SVD}}(q)$. Liu et al. (2025) demonstrate that SVD emergence and Stiefel-based EI maximization are concordant under Gaussian assumptions. In our empirical rolling analysis, the two frameworks exhibit strong positive correlation ($\rho = 0.819$, Spearman $\rho_S = 0.844$; exact dimensional agreement 63.2%, and 87.7% within ± 1 dimension), corroborating the numerical reliability of the Stiefel optimizer.

17 Computational Environment and Software Manifest

All estimations were performed in the following computational environment:

- **High-Performance Hardware:** Multi-threaded host architecture (16 physical cores, 32 threads) and hardware-accelerated tensor execution environments.
- **Operating Systems:** Linux 6.8 (production server) and macOS (Darwin 24.3.0, Apple Silicon reference-parity platform).
- **Python & Core Toolchain:** Python 3.11.0, PyTorch 2.3.1 (FP64 ATen Riemannian tensor engine), NumPy 2.0.2, SciPy 1.17.1, pandas 2.3.3, statsmodels 0.14.6, scikit-learn 1.8.0.
- **Parallel Architecture:** Concurrent multi-process execution partitioning rolling windows and surrogate null regimes across dedicated compute streams.
- **Reproducibility:** Deterministic host-generated orthogonal initialization seeds; CPU and GPU tensor implementations exhibited numerical agreement to at least eight decimal places under the fixed deterministic test configuration.

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